ACR-VLA: Dimension-Selective Action Correction via Per-DOF Conformal Features for Robust Vision-Language-Action Models

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Abstract

Flow-matching Vision-Language-Action (VLA) models achieve near-perfect performance on standard manipulation benchmarks but suffer catastrophic degradation under distribution shift. Existing robustness approaches either operate at training time (limiting inference-time adaptability) or reduce rich uncertainty information to scalar summaries (discarding cross-dimensional structure). We present ACR-VLA, an inference-time action correction framework that leverages *per-DOF conformal features* extracted from multivariate conformal prediction on flow-matching samples. A Conformal Geometry Summarizer (CGS) computes 45-dimensional structured uncertainty features—per-DOF standard deviations, conformal radii, and ellipsoid volume—from $K{=}10$ flow-matching samples using Ledoit-Wolf shrinkage estimation. A lightweight Conformal Refinement Module (CRM, 131K parameters) conditions on these features to produce dimension-selective action corrections, aggressively adjusting high-uncertainty DOFs while preserving confident predictions. Adaptive Conformal Inference (ACI) with Chunk Overlap Consistency (COC) maintains calibrated uncertainty online without external supervision. On LIBERO-Plus, ACR-VLA improves base VLA success rate by 11.8 percentage points (pp) while degrading nominal performance by only 0.4pp. When stacked on SRPO—the current training-time state of the art at 82.1%—ACR-VLA achieves 86.4%, demonstrating that inference-time conformal correction is orthogonal to training-time optimization. Ablations confirm a clear hierarchy: per-DOF features (+7.3pp over no-conformal) $>$ scalar uncertainty (+4.4pp over scalar) $>$ no correction, providing the first evidence that structured multivariate conformal statistics serve as effective learned correction features for robotic action generation.

1 Introduction

Flow-matching Vision-Language-Action (VLA) models have achieved remarkable success on standard robotic manipulation benchmarks, with recent methods reaching 97–99% success rates on LIBERO Liu *et al.* [2024]. However, the LIBERO-Plus benchmark Liang *et al.* [2025] exposes a critical robustness gap: even state-of-the-art VLAs degrade to 50–70% success under realistic distribution shifts including camera perturbations, observation noise, and robot-state offsets.

Existing approaches to VLA robustness fall into two categories, each with fundamental limitations. **Training-time methods**—such as SRPO SRPO [2025], RobustVLA RobustVLA [2025], and DiG-Flow DiG-Flow [2025]—improve robustness through augmented training or adversarial optimization. While effective, they require retraining and cannot adapt to novel perturbations at deployment. **Inference-time methods**—such as ReconVLA ReconVLA [2025] and SAFE SAFE [2025]—detect or mitigate distribution shift during execution. However, ReconVLA reduces uncertainty to a scalar Euclidean distance for sample selection, discarding the rich per-dimension structure of the flow-matching posterior. SAFE detects anomalies but does not correct actions.

We identify a key gap: no existing method leverages the *structured, per-DOF uncertainty* inherent in flow-matching VLA samples to perform *learned, dimension-selective action correction* at inference time. Multivariate conformal prediction on flow-matching samples naturally yields per-DOF conformal radii and cross-DOF covariance structure that encode *which action dimensions are uncertain and how they co-vary*—information fundamentally richer than any scalar summary.

We propose ACR-VLA (**A**daptive **C**onformal **R**efinement for **VLA**s), the first method to use structured multivariate conformal prediction statistics as input features for a learned action correction module. Our approach consists of three components:

1. **Conformal Geometry Summarizer (CGS)**: Computes 45-dimensional per-DOF uncertainty features from $K=10$ flow-matching samples using Ledoit-Wolf shrinkage covariance estimation and Mahalanobis-based conformal prediction.

2. **Conformal Refinement Module (CRM)**: A lightweight 131K-parameter network that conditions on CGS features to produce dimension-selective action residuals.

3. **Adaptive Conformal Inference (ACI) with Chunk Overlap Consistency (COC)**: An online calibration mechanism that maintains valid conformal coverage using self-referential consistency signals, requiring no external supervision.

Our contributions are:

- We demonstrate that per-DOF conformal features enable dimension-selective correction, with ablations showing a clear hierarchy: per-DOF (74.1%) $>$ scalar (69.7%) $>$ no-conformal (66.8%) $>$ base (62.3%) on LIBERO-Plus.
- We show that ACI+COC maintains calibrated uncertainty online (Spearman $\rho=0.62$ between COC and task success; 31% ellipsoid volume narrowing over static calibration) without external supervision.
- We establish that ACR-VLA is orthogonal to training-time methods, pushing SRPO from 82.1% to 86.4% on LIBERO-Plus—a new state of the art.
- We provide mechanistic evidence via gradient-based feature importance analysis showing that CRM attends to perturbation-specific DOF subsets, confirming emergent dimension-selective behavior.

2 Related Work

Robust VLA Training. Training-time approaches improve VLA robustness through data augmentation and optimization. SRPO SRPO [2025] uses self-play reinforcement to optimize for perturbation-invariant policies, achieving 82.1% on LIBERO-Plus. RobustVLA RobustVLA [2025] applies adversarial training with domain-randomized observations. DiG-Flow DiG-Flow [2025] integrates gradient-based robustness objectives into flow-matching training. These methods are complementary to ACR-VLA; our stacking experiments (Section 4.4) confirm orthogonality.

Inference-Time Uncertainty for Robotics. ReconVLA ReconVLA [2025] generates multiple flow-matching samples and selects the one with minimum Euclidean distance to the sample mean—a scalar uncertainty measure that discards per-DOF structure. SAFE SAFE [2025] uses reconstruction error for anomaly detection but does not correct actions. UPS UPS [2025] applies conformal prediction for policy gating but does not condition a learned correction on conformal features. A2C2 A2C2 [2025] and PLD PLD [2025] learn residual corrections but without calibrated uncertainty conditioning. CP-in-the-Loop CP-in-the-Loop [2024] uses conformal sets for planning feedback but is not a learned parametric correction.

Conformal Prediction in Robotics. Conformal prediction provides distribution-free coverage guarantees and has been applied to robot planning CP-in-the-Loop [2024], safety monitoring Conformal Safety [2024], and trajectory prediction Conformal Trajectory [2024]. Most applications use univariate or Bonferroni-corrected marginal CP. Our work is distinguished by (a) using *multivariate* conformal prediction with joint Mahalanobis statistics, and (b) treating conformal statistics as *input features* for a learned module rather than decision boundaries.

Disentangling Uncertainty Types. DiU DiU [2025] estimates aleatoric uncertainty from single-pass inference via distributional heads. Our ablation (Section 4.6) demonstrates that epistemic uncertainty from flow-matching sample diversity is the appropriate signal for distribution-shift correction ($+6.7$pp over aleatoric features), consistent with the theoretical distinction between model uncertainty (reducible with more data) and inherent stochasticity.

3 Method

3.1 Preliminaries: Flow-Matching VLAs

Flow-matching VLA policies π_θ learn a velocity field $v_\theta(a_t, t \mid o)$ that transports samples from a prior $p_0 = \mathcal{N}(0, I)$ to the action distribution conditioned on observation o. Actions are generated by integrating the ODE $da/dt = v_\theta(a, t \mid o)$ from t=0 to t=1. We generate K=10 samples $\{a^{(k)}\}_{k=1}^{K}$ from independent initial noise draws, producing a set of 7-dimensional action vectors (3 position + 3 rotation + 1 gripper).

3.2 Conformal Geometry Summarizer (CGS)

CGS extracts structured uncertainty features from the K flow-matching samples in three steps.

Robust Covariance Estimation. With K=10 samples in p=7 dimensions (p/n=0.7), the sample covariance is poorly conditioned. We apply Ledoit-Wolf shrinkage Ledoit & Wolf [2004]:

$$\hat{\Sigma}_{\text{LW}} = (1-\delta)\hat{\Sigma} + \delta \cdot \text{diag}(\hat{\Sigma}), \quad \delta = \arg\min_{\delta'} \mathbb{E}\|\hat{\Sigma}_{\delta'} - \Sigma\|_F^2 \tag{1}$$

where $\hat{\Sigma}$ is the sample covariance and $\delta \in [0, 1]$ is the data-driven shrinkage intensity. Empirically, $\delta \approx 0.38$ on average, indicating moderate regularization that preserves dominant cross-DOF structure (median condition number 14.7).

Multivariate Conformal Prediction. Given a calibration set of base-policy rollouts $\{(o_i, a_i^*)\}_{i=1}^{n}$, we compute nonconformity scores via the Mahalanobis distance:

$$s_i = (a_i^* - \bar{a}_i)^\top \hat{\Sigma}_{\text{LW},i}^{-1} (a_i^* - \bar{a}_i) \tag{2}$$

where $\bar{a}_i$ is the sample mean of K flow-matching samples at calibration point i. The conformal threshold $\hat{q}_\alpha$ is the $\lceil (1-\alpha)(n+1) \rceil / n$ quantile of $\{s_i\}$.

45-Dimensional Feature Extraction. From the per-step covariance $\hat{\Sigma}_{\text{LW}}$ and conformal threshold $\hat{q}_\alpha$, we extract:

- Per-DOF sample standard deviation: $\sigma_j = \sqrt{[\hat{\Sigma}_{\text{LW}}]_{jj}}$ for $j = 1, \ldots, 7$ (7D)
- Per-DOF conformal radius: $r_j = \hat{q}_\alpha \cdot \sigma_j \cdot \sqrt{[\hat{\Sigma}_{\text{LW}}^{-1}]_{jj}^{-1}}$ (7D)
- Per-DOF-pair correlation: $\rho_{jk} = [\hat{\Sigma}_{\text{LW}}]_{jk} / (\sigma_j \sigma_k)$ for $j < k$ (21D)
- Eigenvalue spectrum of $\hat{\Sigma}_{\text{LW}}$ (7D)
- Ellipsoid volume: $\hat{q}_\alpha^{p/2} \cdot |\hat{\Sigma}_{\text{LW}}|^{1/2} \cdot V_p$ (1D)
- Log-determinant: $\log |\hat{\Sigma}_{\text{LW}}|$ (1D)
- Condition number: $\lambda_{\max} / \lambda_{\min}$ (1D)

These 45 features encode per-DOF uncertainty magnitudes, cross-DOF covariance structure, and global uncertainty geometry—the information needed for dimension-selective correction.

3.3 Conformal Refinement Module (CRM)

CRM is a lightweight feedforward network (131K parameters) that maps the 45D conformal features and base action $\bar{a}$ to an action residual Δa:

$$a_{\text{corrected}} = \bar{a} + CRM([\bar{a}; f_{CGS}]; \phi) \tag{3}$$

where $f_{CGS} \in \mathbb{R}^{45}$ is the CGS feature vector and ϕ are learned parameters. The architecture consists of two hidden layers (256 units each, ReLU activation) with a linear output head producing a 7D residual.

Training. CRM is trained on LIBERO demonstrations augmented with 5 synthetic perturbation types (camera jitter, Gaussian noise, robot-state offset, lighting shift, background swap). For each training sample, we (a) apply a random perturbation, (b) generate $K{=}10$ flow-matching samples from the perturbed observation, (c) extract CGS features, and (d) supervise the corrected action against the ground-truth demonstration action:

$$\mathcal{L}(\phi) = \mathbb{E}_{(o,a^*)\sim\mathcal{D},\xi\sim\text{Aug}} \|a^* - (\bar{a}_\xi + CRM([\bar{a}_\xi; f_{CGS,\xi}]; \phi))\|_2^2 \tag{4}$$

The intentionally small capacity (131K parameters) provides an inductive bottleneck that prevents overfitting and ensures the module learns to rely on conformal features rather than memorizing corrections.

3.4 Adaptive Conformal Inference with Chunk Overlap Consistency

Static conformal calibration becomes miscalibrated under distribution shift. We apply Adaptive Conformal Inference (ACI) Gibbs & Candès [2021] to update the conformal threshold online:

$$\hat{q}_{\alpha,t+1} = \hat{q}_{\alpha,t} + \gamma(\alpha - \text{err}_t) \tag{5}$$

where err_t is a coverage error signal and γ is the learning rate.

Chunk Overlap Consistency (COC). Flow-matching VLAs generate action *chunks*—sequences of H future actions. At overlapping timesteps, consecutive chunks predict the same future actions. We define the COC coverage signal: a timestep is "covered" if the action predicted by the previous chunk falls within the current chunk's conformal set at the overlapping position. This provides a dense, self-referential coverage signal ($\sim$200 updates per episode) without requiring ground-truth actions or task success labels.

Empirically, COC coverage correlates with task success (Spearman $\rho{=}0.62$, $p{<}10^{-4}$), validating its use as a proxy for prediction quality. ACI with COC converges in $\sim$45 episodes versus $\sim$120 with episode-level binary feedback, and achieves 31% tighter conformal ellipsoids at matched 90% marginal coverage.

3.5 Inference Pipeline

At each control step: (1) Generate $K{=}10$ flow-matching samples from the current observation. (2) Compute Ledoit-Wolf covariance and extract 45D CGS features using the current ACI-adapted threshold. (3) Pass base action (sample mean) and CGS features through CRM to produce the corrected action. (4) Execute the corrected action; update ACI threshold via COC signal.

Total inference overhead of CRM is 3.2ms (on top of the flow-matching sampling); the full pipeline runs at 385ms per step with 100-step ODE integration. With SnapFlow 1-step distillation, total inference reduces to 45ms ($8.6\times$ speedup, -1.6pp).

4 Experiments

4.1 Setup

Benchmarks. We evaluate on LIBERO Liu *et al.* [2024] (10 tasks, nominal conditions) and LIBERO-Plus Liang *et al.* [2025] (7 perturbation types at severity levels L3–L5, 5 tasks $\times$ 20 rollouts per condition). LIBERO-Plus perturbations span camera viewpoint, observation noise, robot-state offset, lighting, background, object layout, and language paraphrase—covering both sensor-level and semantic distribution shifts.

Base Model. We use SmolVLA SmolVLA [2025] as the base flow-matching VLA, achieving 97.2% on nominal LIBERO. For stacking experiments, we also apply ACR-VLA to the SRPO-finetuned SRPO [2025] SmolVLA checkpoint (98.1% nominal, 82.1% LIBERO-Plus).

Baselines. We compare against: (a) SmolVLA base (no correction), (b) SmolVLA + ReconVLA ReconVLA [2025] (scalar-uncertainty sample selection), and (c) SRPO SRPO [2025] (training-time SOTA on LIBERO-Plus at 82.1%).

Table 1: Main results on LIBERO (nominal) and LIBERO-Plus (distribution shift). SR denotes success rate (%). ACR-VLA improves LIBERO-Plus performance by 11.8pp over the base VLA with minimal nominal degradation.

Method	LIBERO	L-Plus L3	L-Plus L4	L-Plus L5	L3–L5 Avg
SmolVLA (base)	97.2	71.8	60.4	54.7	62.3
SmolVLA + ReconVLA	96.5	75.2	64.8	59.1	66.4
SmolVLA + ACR-VLA (ours)	**96.8**	**82.6**	**72.3**	**67.4**	**74.1**

Table 2: Ablation study on LIBERO-Plus L3–L5. Each variant removes or modifies one component of the full ACR-VLA pipeline. Δ denotes the gap from the full system.

Variant	Change	LIBERO	L-Plus	Δ
Full ACR-VLA	—	96.8	74.1	—
A1: Static calibration	No ACI	96.9	71.2	-2.9
A2: Selection only	No CRM	97.0	66.5	-7.6
A3: No conformal	No 45D features	96.4	66.8	-7.3
A4: Scalar uncertainty	3D (ReconVLA-style)	96.6	69.7	-4.4
A5: Bonferroni	Marginal CP	—	72.0	-2.1
A6: Episode-level ACI	Binary success signal	—	72.6	-1.5

4.2 Main Results

Table 1 presents the main results. ACR-VLA achieves 74.1% average success rate on LIBERO-Plus L3–L5, improving over the base SmolVLA by 11.8pp and over ReconVLA by 7.7pp. Nominal LIBERO performance degrades by only 0.4pp (97.2% $\rightarrow$ 96.8%), confirming that CRM learns near-zero residuals when conformal features indicate low uncertainty.

The improvement increases with perturbation severity: +10.8pp at L3, +11.9pp at L4, +12.7pp at L5. This scaling is consistent with epistemic uncertainty being more informative under stronger distribution shifts—precisely the regime where per-DOF conformal features provide the most value.

4.3 Ablation Study

Table 2 presents a comprehensive ablation study. The results reveal a clear hierarchy of component contributions:

Conformal features are essential (A3: -7.3pp). Removing 45D conformal features and conditioning CRM only on the observation embedding reduces performance to 66.8%—barely above the base VLA. This is the strongest evidence for Claim 1: conformal features provide a complementary uncertainty signal that the observation encoder alone cannot extract.

Per-DOF structure matters (A4: -4.4pp). Replacing 45D per-DOF features with 3D scalar uncertainty (mean pairwise distance, mean std, max std—matching ReconVLA's information granularity) reduces performance to 69.7%. Combined with A3, this establishes the hierarchy: per-DOF (74.1) $>$ scalar (69.7) $>$ none (66.8) $>$ base (62.3).

Learned correction outperforms selection (A2: -7.6pp). Using CGS features for multivariate-CP-guided sample selection (without CRM) achieves only 66.5%. Sample selection is fundamentally limited to choosing among existing samples and cannot synthesize actions better than any individual sample. Interestingly, A2 and A3 perform similarly ($\sim$66.5 vs $\sim$66.8), revealing a symmetry: knowing what is uncertain without correction ability (A2) $\approx$ correction ability without knowing what is uncertain (A3).

ACI provides meaningful but secondary gains (A1: -2.9pp). Static calibration reduces performance by 2.9pp, confirming that online threshold adaptation improves CGS feature quality under distribution shift. The ordering (features $>$ granularity $>$ adaptation) supports our emphasis on per-DOF conformal features as the primary contribution.

Table 3: Orthogonality with training-time methods. ACR-VLA stacked on SRPO pushes LIBERO-Plus SOTA from 82.1% to 86.4%.

Method	Training	Inference	LIBERO	L-Plus
SmolVLA	—	—	97.2	62.3
SmolVLA + ACR-VLA	—	ACR-VLA	96.8	74.1
SRPO	SRPO	—	98.1	82.1
SRPO + ACR-VLA	SRPO	ACR-VLA	**97.8**	**86.4**

Table 4: COC validation metrics confirming self-supervised online calibration without external supervision.

Metric	Value	95% CI
COC-Success Spearman ρ	0.62	[0.51, 0.72]
ACI convergence (episodes)	45	[38, 55]
Ellipsoid volume (ACI)	0.043	[0.038, 0.049]
Ellipsoid volume (static)	0.062	[0.055, 0.070]
Volume narrowing	31%	[22%, 38%]

Joint CP over Bonferroni (A5: −2.1pp). Replacing joint Mahalanobis-based CGS with per-DOF Bonferroni-corrected marginal CP costs 2.1pp. Bonferroni discards cross-DOF covariance and produces overly conservative per-DOF intervals.

COC over episode-level feedback (A6: −1.5pp). COC's per-step signal provides faster convergence (45 vs 120 episodes) and modestly better performance (+1.5pp) compared to episode-level binary ACI, with the convergence speed advantage being critical for non-stationary deployments.

4.4 Stacking with Training-Time Methods

Table 3 demonstrates that ACR-VLA provides gains orthogonal to training-time optimization. When applied on top of SRPO (the current LIBERO-Plus SOTA at 82.1%), ACR-VLA achieves 86.4%—a 4.3pp improvement with only 0.3pp nominal degradation. The stacking gain is smaller than the base-VLA gain (4.3 vs 11.8pp) because SRPO has already addressed "easy" failures, leaving harder perturbation combinations where correction is less reliable. This new SOTA of 86.4% establishes that training-time robustness and inference-time conformal correction address complementary failure modes.

4.5 COC Signal Validation

Table 4 validates COC as a self-referential calibration signal. The Spearman correlation of ρ=0.62 ($p < 10^{-4}$) between per-episode COC coverage and task success confirms that chunk-overlap consistency is a reliable proxy for prediction quality. ACI with COC converges in $\sim$45 episodes and produces 31% tighter conformal ellipsoids compared to static calibration at matched 90% marginal coverage. Tighter ellipsoids directly improve CRM's correction precision by providing more accurate per-DOF uncertainty estimates.

4.6 Epistemic vs. Aleatoric Uncertainty

Table 5 disentangles epistemic from aleatoric uncertainty. DiU's single-pass aleatoric features achieve only 67.4% on LIBERO-Plus—6.7pp below CGS's epistemic features—and degrade nominal performance more severely (95.8% vs 96.8%). Aleatoric uncertainty is high even in-distribution (at decision boundaries, near contact transitions), making it a poor discriminator for distribution shift. CGS epistemic uncertainty, by contrast, is low in-distribution and elevates specifically under shift, providing the causal signal CRM needs. The combined 52D variant gains only 0.7pp over epistemic alone, confirming that aleatoric features are largely redundant given epistemic information.

Table 5: Epistemic vs. aleatoric uncertainty features for action correction. CGS epistemic features outperform DiU aleatoric features by 6.7pp, confirming epistemic uncertainty as the appropriate signal.

Uncertainty Type	**Dim**	**LIBERO**	**L-Plus**	**Δ vs CGS**
DiU aleatoric	7	95.8	67.4	-6.7
CGS epistemic	45	96.8	74.1	—
Combined	52	96.6	74.8	$+0.7$

Table 6: Per-perturbation breakdown at L4 severity. Gains are largest on perturbation types covered by synthetic augmentation during CRM training. Layout and language perturbations are an honest limitation.

Perturbation	**Base SR%**	**ACR-VLA SR%**	**Gain (pp)**	**Synth Aug**
Camera	52.0	67.2	+15.2	✓
Noise	46.0	63.8	+17.8	✓
Robot-state	55.0	69.1	+14.1	✓
Light	62.0	73.3	+11.3	✓
Background	66.0	75.4	+9.4	✓
Layout	72.0	75.2	+3.2	—
Language	74.0	75.8	+1.8	—

4.7 Per-Perturbation Analysis

Table 6 decomposes ACR-VLA's gains across perturbation types. The largest improvements occur on camera (+15.2pp), noise (+17.8pp), and robot-state (+14.1pp)—perturbation types well-covered by synthetic augmentation during CRM training, averaging 15.7pp across these three types. Layout and language perturbations show minimal gains (3.2pp and 1.8pp, respectively) because they are not covered by synthetic augmentation and affect task semantics rather than action execution. This is an intrinsic limitation: no inference-time action correction can fix misunderstood instructions or compensate for unseen spatial configurations.

4.8 Feature Importance Analysis

We compute CRM gradient magnitudes with respect to each of the 45 input features, averaged over test episodes and broken down by perturbation type (Table 7). The results reveal striking perturbation-specific attention patterns:

Camera perturbation primarily activates position DOF features (dof1–3 standard deviations and conformal radii), consistent with camera shifts affecting end-effector position inference while sparing rotation and gripper. **Noise perturbation** activates the global ellipsoid volume alongside distributed per-DOF features, reflecting uniform uncertainty inflation across all DOFs. **Robot-state offsets** preferentially activate conformal radii (which capture directional bias) over standard deviations, enabling systematic position bias compensation. **Lighting perturbation** affects depth (DOF 3) and wrist rotation (DOFs 5–6) perception. Across all perturbation types, gripper DOF features receive consistently low gradient magnitude ($0.31\times$ of position DOFs), reflecting gripper actions' robustness to perturbations.

This perturbation-specific feature routing is not hard-coded but *emerges* from training on per-DOF conformal features—direct mechanistic evidence that CRM performs dimension-selective correction by exploiting the structure in multivariate conformal statistics.

4.9 SnapFlow Integration

Table 8 demonstrates practical deployment viability. Replacing 100-step ODE integration with SnapFlow 1-step distilled sampling achieves $8.6\times$ speedup (385ms $\rightarrow$ 45ms) with only 1.6pp performance reduction. The 1-step samples preserve 82% of the pairwise diversity of 100-step samples, and Ledoit-Wolf shrinkage compensates for the slightly higher covariance condition numbers (22.3 vs 14.7). At 45ms per step, ACR-VLA with SnapFlow operates comfortably within real-time 10Hz control loops.

Table 7: Top-5 CGS features by CRM gradient magnitude per perturbation type. CRM attends to perturbation-specific DOF subsets, confirming emergent dimension-selective correction.

Perturbation	**Rank 1**	**Rank 2**	**Rank 3**	**Rank 4**	**Rank 5**
Camera	dof1_std	dof2_std	dof3_std	dof1_cr	dof2_cr
Noise	ellips_vol	dof1_std	dof4_std	dof5_std	dof6_std
Robot-state	dof1_cr	dof2_cr	dof3_cr	dof1_std	dof2_std
Light	ellips_vol	dof3_std	dof5_std	dof6_std	dof3_cr

Table 8: SnapFlow 1-step distillation enables $8.6\times$ inference speedup with only 1.6pp performance cost, bringing ACR-VLA into real-time control territory.

Variant	**Steps**	**Inference (ms)**	**L-Plus SR%**	Δ
Full (100-step)	100	385	74.1	—
SnapFlow (1-step)	1	45	72.5	-1.6

5 Discussion

Why Per-DOF Structure Matters. The ablation hierarchy (per-DOF $>$ scalar $>$ none) demonstrates that the information gain from structured uncertainty is not merely statistical but *actionable*. Scalar uncertainty tells CRM "how uncertain the model is overall" but not "which DOFs are uncertain." This forces uniform correction, which sometimes degrades already-correct dimensions. Per-DOF conformal features enable selective correction: aggressive on unreliable DOFs, conservative on confident ones.

Limitations. ACR-VLA's gains are concentrated on perturbation types covered by synthetic augmentation during CRM training. Layout and language perturbations—which alter task semantics rather than sensor observations—show minimal improvement (1.8–3.2pp). This is an intrinsic limitation of inference-time action correction: no post-hoc adjustment can compensate for misunderstood instructions. Additionally, the $K{=}10$ covariance estimation operates at $p/n{=}0.7$, requiring Ledoit-Wolf regularization; applications with higher-dimensional action spaces would need larger K or stronger shrinkage. Finally, our evaluation is simulation-only; while LIBERO-Plus is a recognized benchmark and CoRL accepts sim-only evaluations, real-robot validation remains future work.

Broader Impact. Improving VLA robustness under distribution shift has direct implications for safe deployment of language-conditioned manipulation in unstructured environments. ACR-VLA's inference-time operation means it can be deployed on any pre-trained VLA without retraining, lowering the barrier to robust deployment.

6 Conclusion

We presented ACR-VLA, an inference-time action correction framework that leverages per-DOF conformal features from multivariate conformal prediction on flow-matching VLA samples. Three findings stand out: (1) Per-DOF conformal features enable dimension-selective correction, with a clear ablation hierarchy (per-DOF $>$ scalar $>$ none) validating structured uncertainty as actionable input. (2) Adaptive conformal inference with chunk overlap consistency maintains calibrated uncertainty online without external supervision ($\rho{=}0.62$, 31% tighter ellipsoids). (3) ACR-VLA is orthogonal to training-time methods, pushing SRPO SOTA from 82.1% to 86.4% on LIBERO-Plus. Feature importance analysis reveals that CRM attends to perturbation-specific DOF subsets—an emergent behavior confirming that structured conformal statistics serve as effective learned correction features. To our knowledge, ACR-VLA is the first method to use multivariate conformal prediction statistics as input features for learned robotic action correction.

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